

# MH2500 Probability and Introduction to Statistics

## Slides for Week 12-2

Topics: LLN and CTL

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# Chapter 8

We will discuss in this chapter two famous theorems in probability, the [law of large numbers](#), and the [central limit theorem](#).

- The law of large numbers, essentially says that when an experiment is repeated many times, the proportion of experiments with successful outcomes is approximately the probability of success.
- The central limit theorem quantifies how close the proportion is to the probability of success.

## Markov's inequality

Let  $X$  be a *nonnegative* random variable, then

$$P\{X \geq t\} \leq \frac{\mathbb{E}(X)}{t} \quad \text{for any } t > 0.$$

**Proof:** By

$$\mathbf{1}_{\{X \geq t\}} \leq \frac{X}{t},$$

taking expectation on both sides, we have

$$P\{X \geq t\} = \mathbb{E}(\mathbf{1}_{\{X \geq t\}}) \leq \mathbb{E}\left(\frac{X}{t}\right) = \frac{\mathbb{E}(X)}{t}.$$

## Chebyshev's Inequality

Let  $X$  be a random variable such that  $\mathbb{E}(X)$  and  $\text{Var}(X)$  exist. Then

$$P\{|X - \mathbb{E}(X)| \geq t\} \leq \frac{\text{Var}(X)}{t^2} \quad \text{for any } t > 0.$$

**Proof:**

$$P\{|X - \mathbb{E}(X)| \geq t\} = P\{|X - \mathbb{E}(X)|^2 \geq t^2\}$$

$$\leq \frac{\mathbb{E}(|X - \mathbb{E}(X)|^2)}{t^2}$$

$$= \frac{\text{Var}(X)}{t^2}$$

# Consequences

- Let  $\sigma = \sqrt{\text{Var}(X)}$  and set  $t = k\sigma$ ,

$$P\{|X - \mu| \geq k\sigma\} \leq \frac{\sigma^2}{(k\sigma)^2} = \frac{1}{k^2}.$$

E.g., for  $k = 2$ ,  $P\{|X - \mu| \geq 2\sigma\} \leq \frac{1}{4}$ .

- This is far from the tightness for many particular distributions: For a normal variable  $X$ ,  $P\{|X - \mu| \geq 2\sigma\} \approx 1/20$ .
- But **Chebyshev's inequality is tight**. Consider  $X$  such that

$$P\{X = k\} = P\{X = -k\} = \frac{1}{2k^2}, \quad P\{X = 0\} = 1 - \frac{1}{k^2}.$$

## Corollary

If  $\text{Var}(X) = 0$ , then  $P\{X = \mu\} = 1$ .

**Proof:** Observe that

$$\{|X - \mu| > 0\} = \bigcup_{n=1}^{\infty} \left\{ |X - \mu| \geq \frac{1}{n} \right\}.$$

By Chebyshev's inequality, for any  $\epsilon > 0$ ,

$$P\{|X - \mu| \geq \epsilon\} \leq \frac{\text{Var}(X)}{\epsilon^2} = 0.$$

Hence

$$P\{|X - \mu| > 0\} \leq \sum_{n=1}^{\infty} P\left\{ |X - \mu| \geq \frac{1}{n} \right\} = 0.$$

- How can we get the first inequality?

# The Law of Large Numbers

- 1 If we toss a fair coin many times, we expect about half of them are heads.
- 2 While detained as a prisoner during World War II, South African mathematician [John Kerrich](#) tossed a coin 10,000 times. He obtained 5067 heads.
- 3 Mathematically, the law of large numbers tells us that happens.
- 4 The coin tosses are random independent events. Let  $X_i$  be the indicator variable of obtaining a head on the  $i$ -th toss. Then the  $X_i$ 's are independent.

The proportion of heads is then given by  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ . The law of large numbers tells us that  $\bar{X}_n$  approaches  $\frac{1}{2}$ .

In the case of a fair coin toss, the  $X_i$ 's are Bernoulli random variables with  $p = 1/2$ ,

$$\mathbb{E}(X_i) = \frac{1}{2} \qquad \text{Var}(X_i) = \frac{1}{4},$$

and for  $n = 10000$ ,

$$\text{Var}(\bar{X}_{10,000}) = 2.5 \times 10^{-5}.$$

Hence the standard deviation is 0.005 .

Kerrich's observation was 0.5067. This is 0.0067 away from the mean, which is slightly more than one standard deviation away.

This is consistent with Chebyshev's inequality in the form

$$P\{|\bar{X}_n - \mu| > k\sigma\} \leq 1/k^2.$$



## Weak Law of Large Numbers

Let  $X_1, X_2, \dots, X_i, \dots$  be a sequence of i.i.d. variables with  $\mathbb{E}(X_i) = \mu$  and  $\text{Var}(X_i) = \sigma^2$ . Let  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ . Then, for any  $\epsilon > 0$ ,

$$\lim_{n \rightarrow \infty} P\{|\bar{X}_n - \mu| > \epsilon\} = 0.$$

**Proof:** We have calculated before that

$$\mathbb{E}(\bar{X}_n) = \mu \quad \text{and} \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

The desired result now follows immediately from Chebyshev's inequality, which states that

$$P\{|\bar{X}_n - \mu| > \epsilon\} \leq \frac{\text{Var}(\bar{X}_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \rightarrow 0$$

when  $n \rightarrow \infty$ .

Weak because the probability of “large deviation from mean” goes to 0, but large deviation can still happen.

In fact, a stronger statement holds: “Large deviation” can never happen once  $n$  is large enough.

## Strong Law of Large Numbers

Let  $X_1, X_2, \dots, X_i, \dots$  be a sequence of i.i.d. variables with  $\mathbb{E}(X_i) = \mu$  and  $\text{Var}(X_i) = \sigma^2$ . Let  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ . Then

$$P \left\{ \lim_{n \rightarrow \infty} \bar{X}_n = \mu \right\} = 1.$$

This followed from the fact that

$$\text{Var} \left( \frac{S_n}{n} \right) = \frac{1}{n^2} \text{Var}(S_n) = \frac{1}{n^2} (n\sigma^2) = \frac{\sigma^2}{n} \rightarrow 0.$$

The law of large numbers says that  $S_n/n$  converges to  $\mu$  in probability.

# Central Limit Theorem

The central limit theorem is concerned not with the fact that the ratio  $S_n/n$  converges to  $\mu$  but with **how it fluctuates around  $\mu$** .

- Normalizing the  $S_n$ 's, set

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}.$$

The central limit theorem states that  $Z_n$  converges to the standard normal.

## Central Limit Theorem

Let  $X_1, X_2, \dots$  be a sequence of i.i.d. random variables having mean  $\mu$  and variance  $\sigma^2$ .

- Also assume that their moment-generating function  $M$  is defined in a neighbourhood of zero.

Let

$$S_n = \sum_{i=1}^n X_i.$$

Then

$$\lim_{n \rightarrow \infty} P \left\{ \frac{S_n - n\mu}{\sigma\sqrt{n}} \leq x \right\} = \Phi(x), \quad -\infty < x < \infty.$$

## Central Limit Theorem – Example 1

If 10 fair dice are rolled, find the approximate probability that the sum obtained is between 30 and 40, inclusive.

**Solution:** Let  $X_i$  denote the value of the  $i$ -th die.

$$\mathbb{E}(X_i) = \frac{7}{2}, \quad \text{Var}(X_i) = \frac{35}{12}.$$

Let  $X = X_1 + \cdots + X_{10}$ .

$$\begin{aligned} P\{30 \leq X \leq 40\} &= P\{29.5 \leq X \leq 40.5\} \\ &= P\left\{ \frac{29.5 - 35}{\sqrt{350/12}} \leq \frac{X - 35}{\sqrt{350/12}} \leq \frac{40.5 - 35}{\sqrt{350/12}} \right\} \\ &\approx P\{-1.0184 \leq Z \leq 1.0184\} \\ &= 2\Phi(1.0184) - 1 \approx 0.692 \end{aligned}$$

Here  $Z$  is the standard normal distribution.

## Central Limit Theorem – Example 2

Let  $X_1, \dots, X_{10}$  be i.i.d. uniform random variables on  $(0, 1)$ .

Find  $P\left\{\sum_{i=1}^{10} X_i > 6\right\}$ .

**Solution:** As each  $X_i$  is a uniform random variables on  $(0, 1)$ , we have

$$\mathbb{E}(X_i) = \frac{1}{2} \text{ and } \text{Var}(X_i) = \frac{1}{12}.$$

• Let  $X = X_1 + \dots + X_{10}$ . Then

$$P\{X > 6\} = P\left\{\frac{X - 5}{\sqrt{10/12}} > \frac{6 - 5}{\sqrt{10/12}}\right\}$$

$$\approx P\{Z > 1.095\} = 1 - \Phi(1.095) \approx 0.1367$$

Here  $Z$  is the standard normal distribution.

# Continuity Theorem

- Recall that a distribution function is **uniquely determined by its mgf  $M_n$** .
- The following theorem (**proof omitted**) states that this unique determination holds for limits as well.

## Continuity Theorem

- Let  $F_n$  be a sequence of cdf with  $M_n$  the corresponding moment-generating function.
- Let  $F$  be a cdf with the moment-generating function  $M$ .

If  $M_n(t) \rightarrow M(t)$  for all  $t$  in an open interval containing zero, then  $F_n(x) \rightarrow F(x)$  at all continuity points of  $F$ .

# Proof of CTL:

We first assume that  $\mu = 0$  and  $\sigma = 1$ .

- Since the CDF of  $N(0, 1)$  is continuous, it suffices to show that the mgf of  $(X_1 + \cdots + X_n)/\sqrt{n}$  converges to the mgf of  $N(0, 1)$ .

Assume that  $M(t)$ , the mgf of  $X_i$ , exists and is finite. Then

$$M\left(\frac{t}{\sqrt{n}}\right) = \mathbb{E}\left(e^{\frac{t}{\sqrt{n}}X_i}\right).$$

The mgf of  $(X_1 + \cdots + X_n)/\sqrt{n}$  is  $\left[M\left(\frac{t}{\sqrt{n}}\right)\right]^n$ . Let  $L(t) = \log M(t)$ .

Then  $L'(t) = \frac{M'(t)}{M(t)}$  and  $L''(t) = \frac{M(t)M''(t) - [M'(t)]^2}{[M(t)]^2}$ , which gives

- $L(0) = 0$ ,  $L'(0) = M'(0) = \mu = 0$ ,  $L''(0) = M''(0) = \mathbb{E}(X^2) = 1$ .



To prove  $\lim_{n \rightarrow \infty} [M(t/\sqrt{n})]^n \rightarrow e^{t^2/2}$ , we just need to show:

$$\lim_{n \rightarrow \infty} [n \log M(t/\sqrt{n})] \rightarrow t^2/2,$$

i.e.,

$$\lim_{n \rightarrow \infty} [nL(t/\sqrt{n})] \rightarrow t^2/2,$$

which is true:

$$\begin{aligned} \lim_{n \rightarrow \infty} nL(t/\sqrt{n}) &= \lim_{n \rightarrow \infty} \frac{L(t/\sqrt{n})}{n^{-1}} = \lim_{n \rightarrow \infty} \frac{-L'(t/\sqrt{n})n^{-3/2}t}{-2n^{-2}} \\ &= \lim_{n \rightarrow \infty} \left[ \frac{L'(t/\sqrt{n})t}{2n^{-1/2}} \right] = \lim_{n \rightarrow \infty} \left[ \frac{-L''(t/\sqrt{n})n^{-3/2}t^2}{-2n^{-3/2}} \right] \\ &= \lim_{n \rightarrow \infty} \left[ L'' \left( \frac{t}{\sqrt{n}} \right) \frac{t^2}{2} \right] = \frac{t^2}{2}. \end{aligned}$$

In general,

If  $\mu \neq 0$ , or  $\sigma^2 \neq 1$ , then take  $Y_i = (X_i - \mu)/\sigma$ . Then  $\mathbb{E}(Y_i) = 0$  and  $\text{Var}(Y_i) = 1$ .

By what proved just now, we have

$$\lim_{n \rightarrow \infty} P \left\{ \frac{Y_1 + \cdots + Y_n}{\sqrt{n}} \leq x \right\} = \Phi(x), \quad -\infty < x < \infty,$$

which is what wanted

$$\lim_{n \rightarrow \infty} P \left\{ \frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \leq x \right\} = \Phi(x).$$

# Remarks:

- ① We have given one of the simplest version of the central limit theorem. (There are many other versions.)
- ② We assumed the mgf exists, which is a rather strong assumption.